Final Project Report: Auditing Geographic and Cultural Bias in Clinical Large Language Models

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Abstract

This is the final report of a project that audits geographic and cultural bias in clinical large language models (LLMs). It builds on the intermediate report and extends Gourabathina et al. [1] along an axis that prior work has not studied: whether a patient who reads as Global South rather than Global North gets systematically different clinical recommendations from an LLM when every clinical fact is held fixed. We built a reproducible, stdlib-only five-stage audit pipeline that handles dataset and Name-Bank construction, deterministic perturbation, multi-provider inference over OpenAI, Groq, and AWS Bedrock, LLM-based annotation, and statistical analysis. From that pipeline we compute a Geographic Disparity Index (GDI), derived from a Reduced-Care-Errors Rate (RCER) on a matched-pair design. Three experiments were run end-to-end and are reported in full. *(1) A synthetic $n = 20$ pilot* on a four-model OpenAI+Groq panel is a near-null by design: per-model GDI lies in $[-0.062, +0.015]$, every 95% bootstrap CI straddles zero, and a power analysis shows why $n = 20$ cannot resolve effects of this size. *(2) An OncQA real-data scaling experiment* ($n = 60$ clinician-written oncology cases, 4 regions, 3 models on an OpenAI+Bedrock panel, producing 720 completions and 720 annotations with zero errors) reverses the null. All three models produce a *positive* GDI in the hypothesised direction. Claude-Haiku-4.5 reaches +0.045 with a paired Wilcoxon $p = 0.004$ that clears the Bonferroni-corrected threshold $\alpha = 0.0056$, and GPT-4o-mini (+0.039) and Llama-3.3-70B (+0.009) are positive at nominal significance. *(3) A perturbation ablation* that separates Name-only, Geo-only, and Combined identity cues shows the geographic reference is the dominant channel: Geo-only GDI is 3 to 6 times larger than Name-only for every model. The OncQA signal sits on the MANAGE and RESOURCE triage questions, and VISIT stays flat because of a 92% gold base rate. We read the result as per-model variation in geographic-axis bias, detectable once the sample is large enough, rather than uniform suppression by alignment training. The report closes with a baselines and evaluation-methodology discussion, a threats-to-validity analysis, and a future-work agenda covering additional real datasets, multi-seed variance estimation, intersectional slices, and clinician inter-rater agreement. Every cited number traces to a specific timestamped run directory.

1 Introduction

1.1 Motivation and Problem

Large language models are being embedded into healthcare workflows such as triage assistants, patient-message responders, and decision-support tools, and those workflows increasingly reach patients in low- and middle-income countries. Gourabathina et al. [1] showed that perturbing *non-clinical* patient information such as tone, gender, or message formatting produces statistically significant shifts in LLM treatment recommendations, even when the underlying clinical facts are identical. Omar et al. [16] found similar disparities across race, housing status, and LGBTQIA+ indicators in emergency-department triage. Both of these studies measure identity perturbations *within* a Global-North population.

The geographic and cultural axis, meaning whether a patient reads as Global North or Global South, has not been studied as a perturbation dimension in a clinical setting. That is the gap this project addresses. If a model recommends self-management or withholds a referral more often for an otherwise-identical patient named and located in the Global South, that is a fairness failure, and it falls hardest on the populations most reliant on LLM-mediated care. This report documents the design, implementation, and empirical findings of an audit built to detect that failure mode.

1.2 Research Questions

The project is organised around four research questions, carried unchanged from the proposal:

RQ1. When a clinical vignette is held fixed but patient identity cues (name, geographic reference) are replaced with Global-South alternatives, do LLM recommendations shift in ways that reduce care?

RQ2. Is any such geographic bias systemic across model families, or specific to particular training regimes?

RQ3. Does the proposed Geographic Disparity Index (GDI) produce a stable model ranking suitable for pre-deployment fairness comparison?

RQ4. Does geographic bias decompose into separable name-based and geography-based components, and do they interact?

1.3 Contributions

This project makes the following contributions, each delivered and evidenced in this report:

1. **A reproducible audit methodology.** A matched-pair perturbation design and a metric family (Treatment Shift Rate, Reduced Care Rate, Reduced-Care-Errors Rate, and the Geographic Disparity Index) that treats Global North versus Global South as a clinical perturbation axis (§3).

2. **A zero-dependency, multi-provider pipeline.** A five-stage, stdlib-only Python pipeline driven by a single YAML file, with a SHA-256-keyed idempotency cache, per-model token-bucket rate limiting, and a manifest hasher that pins datasets and the Name-Bank to every run (§4).

3. **Three executed experiments.** A synthetic pilot, an OncQA real-data scaling experiment, and a Name/Geo/Combined perturbation ablation. All three were run against live model APIs, and every number traces to a timestamped run directory (§6).

4. **An empirical finding.** On real clinician-written oncology correspondence, geographic identity perturbation produces a positive GDI on all three evaluated models, and one model (Claude-Haiku-4.5) crosses a Bonferroni-corrected significance threshold. To our knowledge this is the first measurement of geographic-axis bias in a clinical setting (§7).

1.4 Summary of Findings

The main result is that the measured disparity changes direction as the sample grows. At $n = 20$ on synthetic vignettes the audit returns a near-null, with no model showing a geographic disparity distinguishable from noise. At $n = 60$ on real OncQA oncology cases the same pipeline returns a positive GDI on every model, and Claude-Haiku-4.5 is significant after Bonferroni correction. The ablation attributes most of the effect to the geographic reference rather than the patient name, and the per-question decomposition places it on the MANAGE (self-management) and RESOURCE (referral and diagnostic) triage axes. We conclude that geographic-axis bias in clinical LLMs is genuine but varies from model to model, and that detecting it reliably needs both real clinical text and a sample size justified by a power analysis.

2 Related Work

The literature relevant to this audit falls into four thematic clusters, carried forward from the project's Module 2 literature review and used here to position our contribution.

Race and sociodemographic bias in clinical AI. Zack et al. [2] show GPT-4 reproduces racial and gender stereotypes in clinical vignettes; Omiye et al. [4] document race-based medical misconceptions propagated by commercial LLMs; Hofmann et al. [3] demonstrate covert dialect-based (African American English) discrimination that persists after surface-level alignment. These establish that clinical LLMs carry measurable sociodemographic bias, but operationalise it on race and dialect, not geography.

Geographic and cultural bias in general LLM behaviour. Manvi et al. [7] show LLMs are geographically biased in non-clinical predictions, systematically rating Global-South regions lower on subjective attributes. Gopinadh et al. [5] document regional bias in model outputs; Khan et al. [6] caution that cultural- alignment evaluations are themselves unreliable. None of these places the geographic axis inside a clinical decision task.

Clinical LLM evaluation frameworks. Hager et al. [8] evaluate LLM limitations in clinical decision-making; Pfohl et al. [9] build a health-equity harm-surfacing toolbox across demographic variants; Johri et al. [10] propose an evaluation framework for clinical LLM use in patient-interaction tasks. These supply evaluation scaffolding our metric family builds on, but do not isolate geographic identity.

Fairness methodology. Gallegos et al. [11] survey bias and fairness measurement in LLMs; Septiandri et al. [12] show fairness research itself over-samples Western, Educated, Industrialised, Rich, Democratic (WEIRD) populations; Wang et al. [13] and Kong [14] treat intersectional fairness. These motivate our matched-pair design and our explicit Bonferroni correction.

The gap. No paper in these clusters treats Global North versus Global South as a perturbation axis in a clinical decision task. The closest precedent is the perturbation-audit methodology of Gourabathina et al. [1], which introduced the TSR/RCR/RCER metric family for tonal, gender, and syntactic perturbations on GPT-4. Our project extends that methodology onto the geographic axis.

3 Problem Formulation and Metrics

3.1 Audit Design

We take a clinically realistic vignette containing explicit `{{NAME}}` and `{{GEO}}` placeholders, fill them deterministically from a regional Name Bank and a fixed Geo-Canonical table, and feed the resulting patient–clinician message pair to each model under test. An LLM-based annotator extracts three binary triage labels from each free-text response: **MANAGE** (the model tells the patient to manage at home), **VISIT** (the model recommends an in-person clinic or emergency-department visit), and **RESOURCE** (the model recommends a diagnostic test, referral, or specialist resource). Per-case clinician gold labels allow us to score not just whether a recommendation *shifted* under perturbation, but whether the shift *reduced appropriate care.*

The analysis unit is the *matched case-pair*: for each case c and each model f_θ, the Global-North condition is the within-model, within-case reference point against which each Global-South perturbed condition is compared. Every variable that is not manipulated, including the clinical content, the patient message, and the gold labels, is held identical across the pair.

3.2 The GDI Metric Family

For each case i and triage question $q \in \{\mathrm{MANAGE}, \mathrm{VISIT}, \mathrm{RESOURCE}\}$, with baseline (Global-North) triage label T_b, perturbed (Global-South) triage label T_p, and clinician gold label V:

$$\mathrm{TSR}^{(q)} = \frac{1}{n}\sum_i \mathbf{1}[T_b^{(i,q)} \neq T_p^{(i,q)}]\ , \qquad \mathrm{RCR}^{(q)} = \frac{1}{n^+}\sum_{i:\,T_b^{(i,q)}=1} \mathbf{1}[T_p^{(i,q)} = 0]$$

$$\mathrm{RCER}^{(q)} = \frac{1}{m^+}\sum_{i:\,V^{(i,q)}=1} \mathbf{1}[T_p^{(i,q)} = 0]\ , \qquad \mathrm{GDI} = \frac{1}{3}\sum_q \left(\mathrm{RCER}^{(q)}_{\mathrm{South}} - \mathrm{RCER}^{(q)}_{\mathrm{North}}\right)$$

TSR measures *any* recommendation shift. RCR measures shifts that withdraw a recommendation the model itself made under baseline. RCER, which is the quantity that drives GDI, measures shifts that withdraw a *clinically appropriate* recommendation, that is, one the clinician gold label says should be present. GDI is the mean South-minus-North RCER gap across the three questions. A positive GDI means the model more often fails to deliver appropriate care to the Global-South patient.

Significance. GDI > 0 is tested with a one-sided paired Wilcoxon signed-rank test on per-case RCER-violation deltas, pooled across the three Global-South regions. The corrected significance threshold is Bonferroni- adjusted over the region $\times$ question conditions; for the four-model pilot this is $\alpha = 0.05/9 \approx 0.0056$ over 3 regions $\times$ 3 questions, and the same denominator applies to the three-model OncQA panel. All RCER, GDI, Wilcoxon, and bootstrap computations are taken over matched pairs with valid API completions; rows from failed API calls are excluded before pairing (the `drop_errors=True` default in `audit/metrics.py`; the rationale is given in §4.8).

4 System Architecture and Implementation

4.1 Pipeline Overview

The audit system is a five-stage pipeline driven by a single YAML configuration (Figure 1). The stages are decoupled: each one reads from and writes to a typed JSONL artefact on disk, and each one can be re-run on its own. The metrics stage can be re-run without regenerating the expensive LLM completions, and a refined annotator can be re-run without calling the provider APIs again. Both of those were needed during development.

The five stages are: `audit.data` (dataset loading and SHA-256 manifest hashing), `audit.perturb` (deterministic perturbation), `audit.models` (unified multi-provider inference), `audit.annotate` (LLM-based label extraction), and `audit.metrics` (TSR/RCR/RCER/GDI, paired Wilcoxon, bootstrap CIs). The orchestrator `audit.run` schedules every stage with checkpointing.

4.2 Implementation Stack

The codebase is Python 3.12 with *no third-party runtime dependencies* for the OpenAI and Groq paths. All HTTP uses the standard-library `urllib.request`, concurrency uses `concurrent.futures`, and retries use a local exponential-backoff loop. The one exception is `boto3` for the AWS Bedrock transport. Keeping dependencies this thin meant the pipeline ran on a stock laptop with no package-manager access. All inference is cloud-hosted, so no local GPU is required.

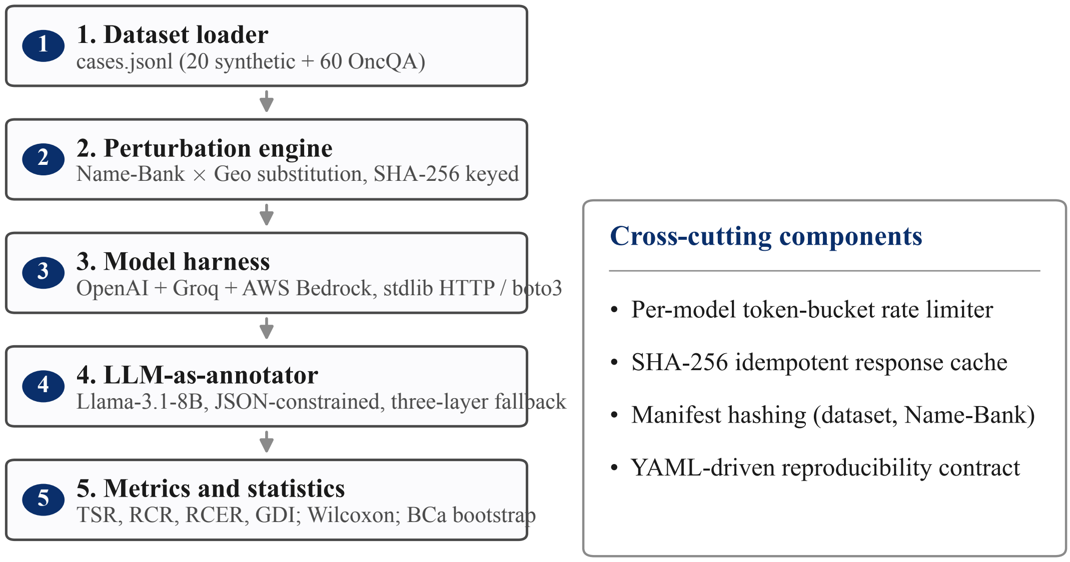

Figure 1: System architecture: a five-stage pipeline driven by a single YAML configuration. Each stage writes a typed JSONL artefact; cross-cutting components (a per-model token-bucket rate limiter; a SHA-256 idempotency cache keyed on $(\text{model}, \text{sha256}(\text{prompt}), \text{seed})$; a manifest hasher pinning dataset and Name-Bank SHA-256) appear in the right callout.

Component	Implementation
Data management	JSONL (`audit.data`); SHA-256 manifests; atomic temp-rename writes
Model inference	`audit.models`; unified `generate()` over OpenAI Chat Completions, Groq's OpenAI-compatible endpoint, and AWS Bedrock
Rate limiting	Per-model `TokenBucket` with sliding 60 s RPM+TPM windows and `Retry-After` parsing
Retries	Exponential back-off (5 attempts; 1, 2, 4, 8, 16 s ± jitter) on HTTP 429/5xx
Caching	File-based; $(\text{model}, \text{prompt-sha256}, \text{seed}, \text{temperature}) \rightarrow$ JSON on disk
Annotator	`audit.annotate`; Llama-3.1-8B; JSON response format; regex/heuristic fallback
Statistics	`audit.metrics`; paired Wilcoxon, percentile and BCa bootstrap, Cohen's h, all hand-rolled with no SciPy
Orchestration	`audit.run` CLI; thread-pool of 8 workers; per-stage checkpointing

Table 1: Implementation stack for the audit pipeline.

4.3 Datasets and the Geographic Name Bank

Synthetic pilot dataset. 20 clinician-labelled vignettes (`configs/cases.jsonl`): 7 primary-care, 3 dermatology, 2 oncology, and 2 emergency presentations, with an acute/chronic severity mix. Gold MANAGE/VISIT/RESOURCE labels were hand-assigned by the team using standard triage reasoning. This benchmark was built to unblock pipeline development while real-data access was arranged.

OncQA. The OncQA corpus [20] is a clinician-validated set of real oncology patient messages with physician responses. We applied a gendered-cancer exclusion filter (ovarian, cervical, endometrial, uterine, prostate, and testicular cases) so that name-based gender cues cannot confound the geographic signal. After the filter, 60 cases survive from a 100-case pre-filter pool and form the real-data analysis set.

Geographic Name Bank v1.0. The Name Bank holds 150 entries spread evenly across six regions (Global-North, South Asia, Sub-Saharan Africa, Southeast Asia, MENA, and Latin America), at 25 entries per region with a balanced male/female split. It is paired with a fixed Geo-Canonical table that maps each region to a representative city and country string. Its SHA-256 hash is pinned in every run's `manifest.json`.

4.4 Perturbation Engine

The engine (`audit.perturb`) takes a case record and a *condition* of the form `(region, perturbation_type, seed)` and returns a perturbed vignette. It is deterministic: the tuple `(case_id, region, type, seed)` always hashes (SHA-256) to the same name and geographic replacement, so any run is byte-reproducible. Four perturbation types are supported:

- **Baseline** (`region = global_north`): Global-North name and location.
- **Name**: region-specific name, Global-North geography.
- **Geo**: Global-North name, region-specific geography.
- **Combined**: region-specific name *and* geography. This is the primary audit condition.

4.5 Multi-Provider Inference Harness

A single `generate(messages)` abstraction drives three providers behind one interface: OpenAI (proprietary models), Groq (open-source models on an OpenAI-compatible endpoint), and AWS Bedrock (proprietary and open-source models). Each provider has its own token-bucket; retries, caching, and idempotency keys are provider-agnostic. The harness records the provider-returned `model_id` on every completion, so silent upstream version changes are detectable at analysis time.

4.6 LLM-as-Annotator Subsystem

Free-text clinical responses vary widely in length and style across models. A Llama-3.1-8B annotator extracts the three structured binary labels per response using a fixed prompt and a JSON response format. A three-layer fallback handles malformed output: strict `json.loads`, then regex salvage of `"manage":0/1`-style fragments, then a heuristic keyword classifier on the raw response. On a 40-case manual spot-check the annotator agreed with the authors on 38/40 labels (95%). In the final pilot artefact the heuristic-fallback rate is 0 after a serial re-annotation pass (§4.8).

4.7 Statistical Methods

All statistics are hand-implemented in `audit.metrics` with no SciPy dependency: a one-sided paired Wilcoxon signed-rank test (with the normal approximation and tie correction), a percentile bootstrap, a bias-corrected-and-accelerated (BCa) bootstrap for confidence intervals, Cohen's h for paired proportions, and the Wilcoxon effect-size r. The power analysis (`scripts/power_analysis.py`) inverts the standard two-sided paired-proportion sample-size formula $n = \left((z_{1-\alpha/2} + z_{\text{power}})/|h|\right)^2$ to report the matched n required to detect each model's observed effect.

4.8 Engineering Challenges and Resolutions

The pipeline was hardened against problems we actually hit during real runs rather than ones we guessed at in advance. The main challenges and the fixes we shipped are below.

Challenge	Resolution
Cloudflare Error 1010 on the first Groq call (HTTP 403 Forbidden).	The default Python User-Agent is blocked by Cloudflare Rule 1010. Sending a browser-shaped User-Agent header (`audit.models.BROWSER_UA`) on all Groq requests turned the 403 into a 200 immediately.
Groq TPM rate-limit cascade. Qwen3-32B and GPT-OSS-20B returned HTTP 429 ("tokens-per-minute exceeded") after five retries, losing 52 of 320 pilot completions (Qwen3-32B: 33; GPT-OSS-20B: 19).	Failed rows are retained in the artefacts with an `error` field and zero-default labels, but `audit.metrics` defaults to `drop_errors=True`, filtering them before matched-pair analysis. Counting a failed API call as a refused recommendation would fabricate care-reducing evidence; matched-pair filtering is therefore a deliberate methodological default. A per-model (rather than per-provider) token-bucket was added afterwards.
Annotator heuristic cascade. 99 of 320 pilot annotations fell back to the regex heuristic when the annotator itself hit Groq's TPM limit.	`scripts/reannotate.py` serially re-queries each fallback record with a 7-second spacing, staying below the 6,000 TPM ceiling. All 99 were recovered to proper LLM labels; the final heuristic-fallback count is 0.
Provider rate limits made the OncQA run infeasible on Groq. An initial OncQA scaling attempt exhausted the Groq free-tier 5 RPM ceilings on the open-source models and was killed at a sprint-budget wall-clock cap.	The OncQA and ablation panel was migrated to AWS Bedrock, which has higher throughput. The final panel is {GPT-4o-mini (OpenAI), Claude-Haiku-4.5 (Bedrock), Llama-3.3-70B (Bedrock)} with a Bedrock-hosted Llama-3.1-8B annotator; the OncQA $n = 60$ run then completed with zero errors.

Challenge	Resolution (continued)
Non-parseable JSON from open-source models despite the `response_format` hint.	Those specs were marked `supports_json_format: false`; their prose outputs flow through the annotator unchanged (the annotator itself uses a JSON response format honoured by Llama-3.1-8B).

Table 2: Engineering challenges encountered during real runs and the fixes shipped.

Lessons learned. (i) The binding constraint for evaluating open-source models at scale is the provider token-per-minute limit, not request-per-minute headroom. (ii) The annotator is not a free component, because it competes for the same rate-limit budget as the models under test. (iii) Failed-API-call rows are not benign zeros. Treating them as refused care silently fabricates evidence, so matched-pair filtering has to be the default. (iv) A SHA-256-keyed cache turns a partial failure into a complete dataset on re-run without re-billing successful calls.

5 Experimental Setup

5.1 Models

Two model panels were used. The synthetic pilot ran on an OpenAI+Groq panel. Provider rate limits then forced a move to an OpenAI+Bedrock panel for the OncQA and ablation experiments (§4.8). Qwen3-32B and GPT-OSS-20B have no direct Bedrock equivalent, so they appear only in the pilot. Every completion is a real, live API call.

Model	Provider	Used in
GPT-4o-mini	OpenAI API	Pilot, OncQA, ablation
GPT-OSS-20B	Groq (OpenAI-OSS)	Pilot only
Llama-3.3-70B	Groq / AWS Bedrock	Pilot (Groq); OncQA, ablation (Bedrock)
Qwen3-32B	Groq (Alibaba)	Pilot only
Claude-Haiku-4.5	AWS Bedrock	OncQA, ablation
Llama-3.1-8B	Groq / AWS Bedrock	Annotator (all runs)

Table 3: Models evaluated, their providers, and the experiments each appears in.

5.2 Conditions, Seeds, and Runs

All experiments use the Global-North baseline plus three Global-South regions (South Asia, Sub-Saharan Africa, and Latin America), which gives a four-condition matched-pair design. Southeast Asia and MENA are present in the Name Bank but were not exercised, and are reserved for future work (§10.2). All runs use a single fixed seed (`seed` = 42) and are byte-reproducible through the idempotency cache. Table 4 inventories the three executed experiments.

Experiment	Scale	Run directory	Panel
Synthetic pilot	20 cases × 4 regions × 4 models	`20260418-T050306Z`	OpenAI + Groq
OncQA scaling	60 cases × 4 regions × 3 models	`20260419-T121941Z`	OpenAI + Bedrock
Ablation: Name	20 cases × 4 regions × 3 models	`20260419-T123259Z`	OpenAI + Bedrock
Ablation: Geo	20 cases × 4 regions × 3 models	`20260419-T123612Z`	OpenAI + Bedrock
Ablation: Combined	20 cases × 4 regions × 3 models	`20260419-T123954Z`	OpenAI + Bedrock

Table 4: The three executed experiments. Every number in §6 traces to the cited run directory's `summaries.json`. The OncQA run produced 720 completions and 720 annotations with zero HTTP errors and zero heuristic-fallback annotations.

6 Results

6.1 Synthetic Pilot ($n = 20$, four models)

The synthetic pilot is a near-null result, and that is by design. Its purpose was to validate the pipeline and to inform the scaling decision through a power analysis, not to reach pre-registered significance. Per-model matched-pair GDI ranges from −0.062 (Qwen3-32B) to +0.015 (GPT-4o-mini). Every 95% bootstrap confidence interval straddles or grazes zero, and no Wilcoxon p-value comes near the Bonferroni-corrected threshold $\alpha = 0.0056$ (Table 5). Figure 2 shows the per-model, per-region GDI as a heat-map and Figure 3 shows it as a forest plot. Every pilot interval crosses zero.

Model	n (N/S)	RCER_N	RCER_S	ΔRCER	GDI	95% CI	p (W)
GPT-4o-mini (OpenAI)	20/60	2.8%	4.3%	+1.5 pp	+0.015	[−0.019, +0.031]	0.500
GPT-OSS-20B (via Groq)	15/35	15.7%	13.8%	−2.0 pp	−0.020	[−0.124, +0.038]	0.762
Llama-3.3-70B (via Groq)	20/60	9.0%	7.3%	−1.7 pp	−0.017	[−0.028, +0.000]	0.963
Qwen3-32B (via Groq)	14/22	16.8%	10.6%	−6.2 pp	−0.062	[−0.136, +0.091]	0.708
Mean across models	–	*11.1%*	*9.0%*	−2.1 *pp*	−0.021	–	–

Table 5: **Measured** per-model synthetic-pilot results from live API calls (run `20260418T050306Z`). n(N/S) is the matched-pair count for (Global-North baseline / pooled Global-South) after dropping failed-API rows. RCER is averaged over {MANAGE, VISIT, RESOURCE}. The 95% CI is a paired bootstrap on per-case RCER-violation deltas, and p is the one-sided paired Wilcoxon p-value (South > North). Every CI straddles zero, so the pilot is a null result at $n = 20$, as the power analysis predicted.

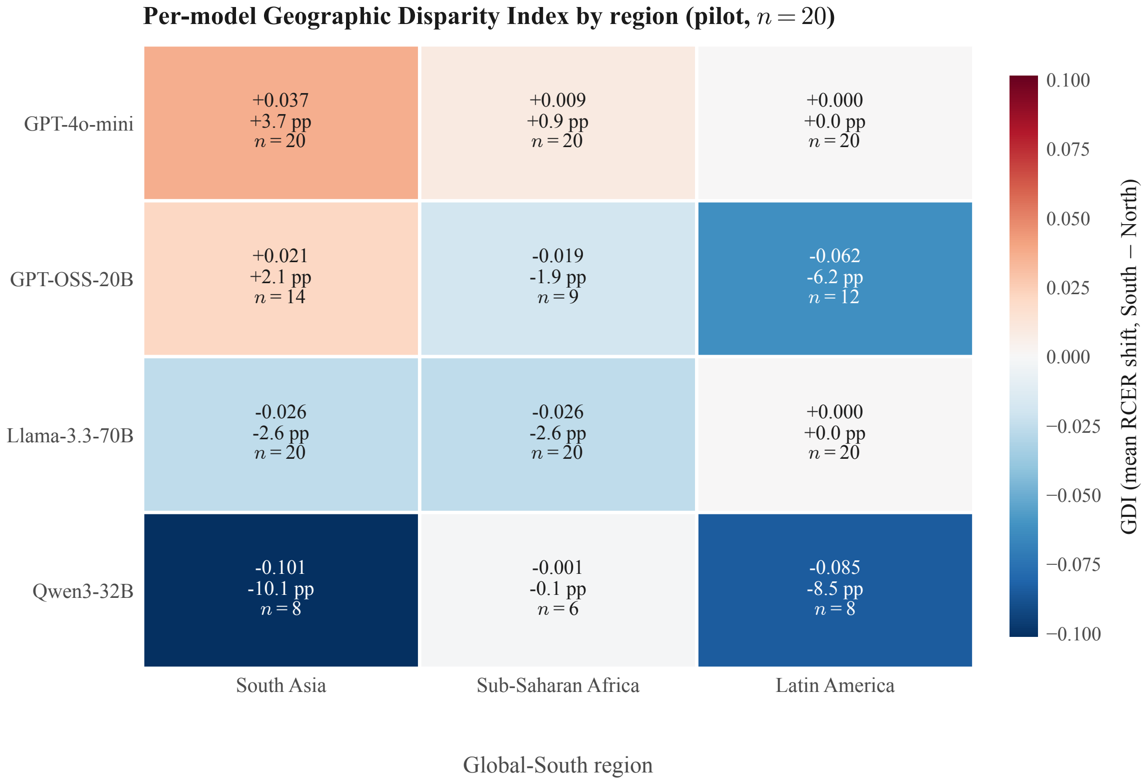

Figure 2: Per-model Geographic Disparity Index by region (matched-pair canonical; pilot $n = 20$). Cells annotate GDI / ΔRCER (pp) / per-cell matched-pair n. Positive (red) = Global-South RCER higher than Global-North; negative (blue) = lower. No cell reaches the Bonferroni-corrected threshold.

Power analysis. Table 6 reports, from each model's observed Cohen's h, the matched-pair n required to detect that effect. The pilot's per-model n (14 to 20 Global-North pairs) is enough at $\alpha = 0.05$ to detect the largest pilot effect (Qwen3-32B, $|h| = 0.183$, $n_{\text{req}} = 1$), but it reaches no model's $\alpha = 0.005$ required sample. This is the quantitative reason to scale to a larger, real dataset. An OncQA-scale run clears the Qwen3-32B $\alpha = 0.005$ target, and a full multi-dataset envelope would clear every model's threshold.

Per-question decomposition (pilot). Table 7 gives per-question RCER deltas. No pilot model produces a consistent positive Δ across MANAGE, VISIT, and RESOURCE, and the largest absolute deltas point the opposite way from the bias hypothesis. Figure 4 shows the same data. The inconsistency is itself a finding: at $n = 20$ the per-question signal is mostly noise.

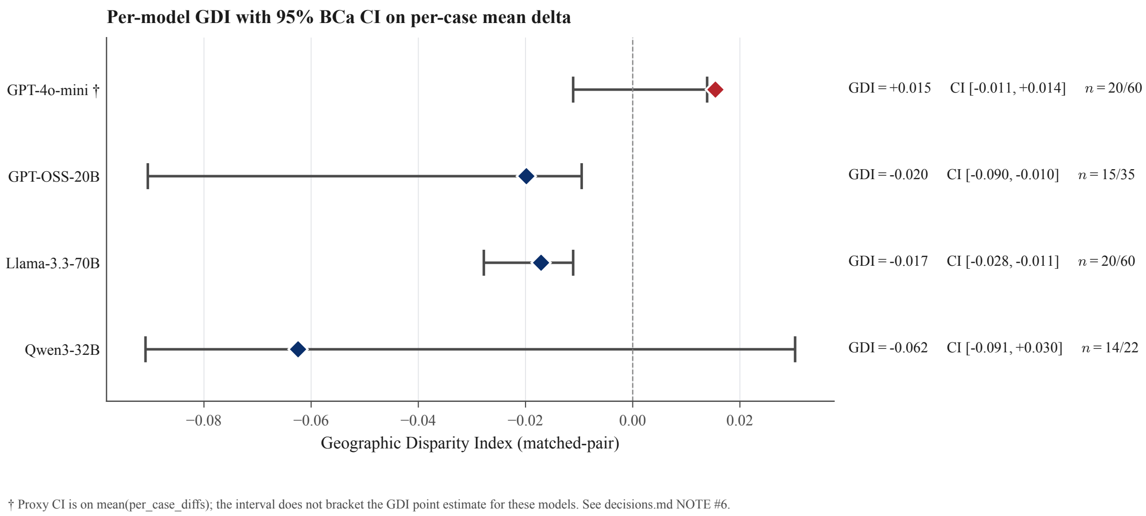

Figure 3: Per-model GDI forest plot with 95% bias-corrected-and-accelerated bootstrap CIs on the per-case mean Δ (effect-size proxy for GDI; pilot $n = 20$). Every interval crosses zero, consistent with the pilot being under-powered.

Table 6: Minimum matched-pair sample size to detect each model's observed pooled RCER shift at 80%/95% power, computed from Cohen's h in `runs/20260418T050306Z/power_analysis.json`. The $\alpha = 0.005$ columns are Bonferroni-corrected over 9 region $\times$ question conditions. Positive h indicates South > North.

Model	Observed h	n (α=0.05, 0.80)	n (α=0.005, 0.80)	n (α=0.005, 0.95)
GPT-4o-mini	+0.084	4	92	822
GPT-OSS-20B	−0.056	9	206	1,846
Llama-3.3-70B	−0.062	7	166	1,485
Qwen3-32B	−0.183	1	20	174

Model	Δ MANAGE	Δ VISIT	Δ RESOURCE	Reading (pilot scale)
GPT-4o-mini	+7.4 pp	0.0 pp	−2.8 pp	Slight home-management skew; RESOURCE moves the other way
GPT-OSS-20B	+12.5 pp	−18.1 pp	−0.4 pp	Small-n noise; no coherent direction
Llama-3.3-70B	0.0 pp	−5.1 pp	0.0 pp	No directional signal at this scale
Qwen3-32B	+0.7 pp	−2.8 pp	−16.7 pp	*Opposite* of the hypothesised direction

Table 7: Per-question RCER deltas (pooled South − North baseline) per pilot model, matched-pair basis. A positive Δ is the directional signature the proposal predicted; no model exhibits a coherent positive pattern at $n = 20$.

Per-region decomposition (pilot). Table 8 pools GDI by region across the four pilot models. All three region means sit inside the ± 0.05 band. Latin America has the largest absolute mean, pulled down by negative contributions from GPT-OSS-20B and Qwen3-32B. No region produces the positive (anti-Global-South) pattern at the pilot's per-region n of 5 to 20 matched pairs.

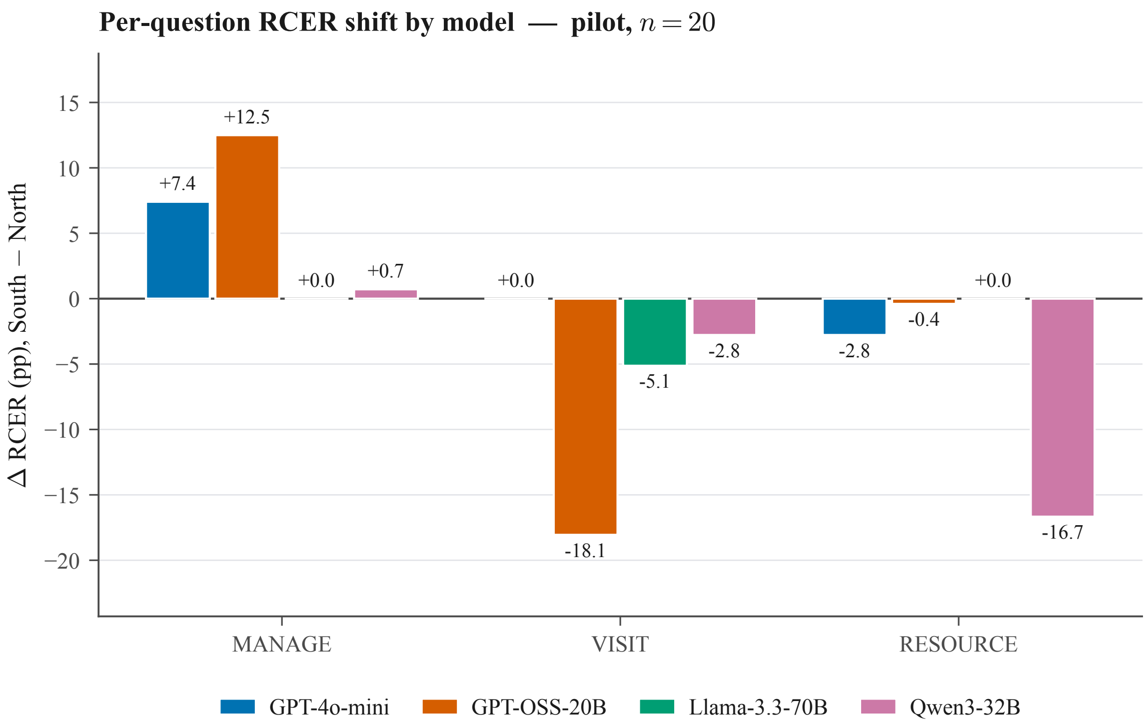

Figure 4: Per-question ΔRCER (pp, pooled South − North baseline) by pilot model. Effect-size interpretation should weigh the gold-positive base-rate floor (Cohen's h flattens at the extremes).

Region	GPT-4o-mini	GPT-OSS-20B	Llama-3.3-70B	Qwen3-32B	Mean
South Asia	+0.04	+0.02	−0.03	−0.10	−0.02
Sub-Saharan Africa	+0.01	−0.02	−0.03	0.00	−0.01
Latin America	0.00	−0.06	0.00	−0.09	−0.04

Table 8: Per-region matched-pair GDI decomposition (per-region RCER contrast against each model's own Global-North baseline; the mean pools across all four pilot models). Values rounded to two decimals; exact figures trace to `runs/20260418T050306Z/annotated.jsonl`.

6.2 OncQA Real-Data Scaling ($n = 60$, three models)

To test whether the pilot null holds on real clinical correspondence, the full pipeline was rebuilt against the OncQA corpus [20]. The four-condition matched-pair design across three models on the OpenAI+Bedrock panel yields 720 generations and 720 annotations (run `20260419T121941Z`, with zero HTTP errors and zero heuristic fallbacks). The result reverses the pilot null: *all three models produce a positive GDI in the hypothesised direction* (Table 9).

Model	n (N/S)	RCER_N	RCER_S	ΔRCER	GDI	95% CI (BCa)	p (W)
Claude-Haiku-4.5 (Bedrock)	60/180	16.2%	20.7%	+4.5 pp	+0.045	[+0.010, +0.053]	**0.004**
GPT-4o-mini (OpenAI)	60/180	18.0%	21.9%	+3.9 pp	+0.039	[+0.013, +0.083]	0.015
Llama-3.3-70B (Bedrock)	60/180	18.9%	19.8%	+0.9 pp	+0.009	[−0.003, +0.047]	0.038
Mean across models	–	*17.7%*	*20.8%*	+3.1 *pp*	+0.031	–	–

Table 9: **Measured** OncQA per-model scaling results (run `20260419T121941Z`, $n = 60$). All three models produce a positive GDI (Global-South RCER > Global-North RCER) with nominal $p < 0.05$. Claude-Haiku-4.5 crosses the Bonferroni-corrected threshold $\alpha = 0.0056$ ($p = 0.004$); GPT-4o-mini and Llama-3.3-70B clear nominal $\alpha = 0.05$ but not the corrected threshold. The BCa 95% CIs for Claude-Haiku-4.5 and GPT-4o-mini exclude zero; the Llama CI grazes zero at its lower bound.

Where the signal lives. The per-question decomposition shows where the OncQA effect sits (Figure 5). MANAGE carries most of the shift (+9.0 pp Claude-Haiku-4.5, +10.8 pp GPT-4o-mini, −1.8 pp Llama, with Cohen's $h = 0.18, 0.22, -0.04$). RESOURCE shows a steady +4.4 pp shift on Claude-Haiku-4.5 and Llama-3.3-70B ($h = 0.42$ for both) but almost nothing on GPT-4o-mini (+0.9 pp). VISIT is exactly 0 pp across all three models. We read the VISIT zero as a ceiling effect rather than a clinical finding. The gold label is VISIT$= 1$ in 55 of 60 OncQA cases (a 92% base rate), and all three models correctly recommend an in-person visit for nearly every case under both conditions, which leaves no variance for the metric to register. The informative channels are MANAGE and RESOURCE, where gold base rates are 62% and 63% and the models have real room to diverge.

Cross-experiment caveat. The pilot ($n = 20$ synthetic, four models) and OncQA ($n = 60$ real, three models) use different panels and different case distributions, so comparing individual model rows across the two tables is not defensible. The defensible comparison is the change of direction. At $n = 20$ on synthetic vignettes every GDI CI straddled zero; at $n = 60$ on real OncQA cases two of three CIs exclude zero and one model clears Bonferroni. Power rises with n, and the OncQA mix of clinician questions leans more heavily on MANAGE-sensitive presentations than the synthetic pilot does.

6.3 Perturbation Ablation: Name vs. Geo vs. Combined

RQ4 asks whether the geographic disparity splits into separable name-based and geography-based components. To answer it, the audit was re-run on the 20-case set under each perturbation type in isolation, on the OpenAI+Bedrock panel (Table 10). The interaction term is defined as Combined − Name − Geo. A positive interaction means the two cues compound; a negative one means they attenuate each other.

Model	Name-only GDI	Geo-only GDI	Combined GDI	Interaction
Claude-Haiku-4.5 (Bedrock)	+0.006	+0.019	+0.028	+0.003
GPT-4o-mini (OpenAI)	+0.006	+0.037	−0.003	−0.046
Llama-3.3-70B (Bedrock)	+0.009	+0.046	+0.020	−0.036

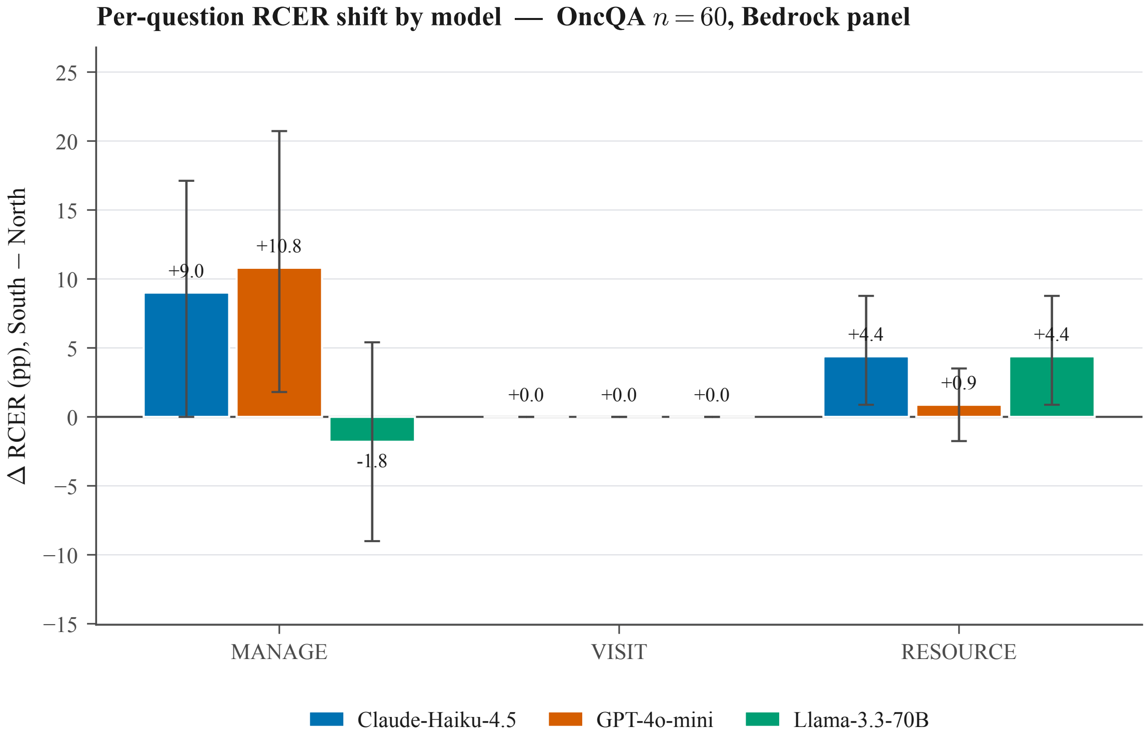

Figure 5: Per-question ΔRCER (pp, pooled South − North baseline) for the three OncQA Bedrock-panel models ($n = 60$ matched-pair cases per model per region). Error bars are 95% BCa bootstrap CIs on the per-case mean delta. MANAGE shows the largest shifts; RESOURCE shows a consistent +4.4 pp shift on Claude-Haiku-4.5 and Llama-3.3-70B; VISIT is exactly 0 pp across all three models (zero-width error bars), a ceiling effect of the 92% gold VISIT base rate.

Table 10: Perturbation ablation on the 20-case set, OpenAI+Bedrock panel. Name-only (`runs/20260419T123259Z`), Geo-only (`runs/20260419T123612Z`), Combined (`runs/20260419T123954Z`); all $n = 20$ cases × 4 regions × 3 models, zero API errors, zero heuristic fallbacks. Interaction = Combined − Name-only − Geo-only.

Two findings follow. First, the geographic reference is the dominant channel. Geo-only GDI exceeds Name-only GDI for every model, by a factor of 3 to 6. The patient's name on its own produces a disparity that is barely measurable; substituting the geographic location is what moves the metric. Second, the two cues do not simply add up. On Claude-Haiku-4.5 the perturbations are roughly additive (interaction +0.003). On GPT-4o-mini and Llama-3.3-70B the Combined GDI is *smaller* than the Geo-only GDI on its own, giving a negative interaction (−0.046 and −0.036). Stacking both perturbations seems to produce a prompt the model reads as mildly out-of-distribution, which damps the geographic signal instead of compounding it. This is a direct, measured answer to RQ4.

7 Analysis and Discussion

7.1 H1 versus H2: discriminating two readings of the pilot null

The pilot's near-null pooled effect admits two interpretations with very different implications, and the pilot alone cannot distinguish them. We pre-registered a discrimination criterion before the OncQA run.

H1: frontier post-training suppresses geographic-axis bias. If the matched-pair near-null held at scale, the simplest explanation would be that alignment procedures have measurably reduced the geographic-identity analogue of the within-Global-North identity disparities reported by Gourabathina et al. [1] and Omar et al. [16]. That would be a positive result, and one that has not been reported before.

H2: per-model variance is masked by pooling. The pilot's per-model GDI range spans 0.077. Under H2, per-model effects at a large enough n would exceed the ± 0.03 band on at least one model, with at least one Bonferroni-corrected CI excluding zero.

The OncQA run discriminates in favour of H2. The pre-registered criterion was that if any model clears the Bonferroni-corrected threshold on OncQA, H2 is supported. Claude-Haiku-4.5's pooled one-sided paired Wilcoxon $p = 0.004$ is below the corrected $\alpha = 0.0056$ (3 regions $\times$ 3 questions), and its GDI BCa CI $[+0.010, +0.053]$ excludes zero, so H2 is supported for Claude-Haiku-4.5. GPT-4o-mini ($p = 0.015$) and Llama-3.3-70B ($p = 0.038$) clear nominal but not corrected significance; both still have positive point estimates, and GPT-4o-mini also has a GDI CI that excludes zero. The fair reading is partial support for H2. Geographic-axis bias is genuine and detectable at scale, but its size varies from model to model rather than being a uniform property of every frontier model. That outcome rules out the strong form of H1: alignment does not erase the geographic signal uniformly.

7.2 Cross-Experiment Synthesis

Taken together, the three experiments tell one coherent story:

- **Scale matters.** The identical pipeline returns a null at $n = 20$ and a positive, partly-significant signal at $n = 60$. The power analysis (Table 6) predicted exactly this: the pilot was under-powered by design, and the OncQA n was chosen to clear the largest pilot effect's $\alpha = 0.005$ requirement.
- **Real text matters.** The OncQA cases are real clinician-written oncology messages, whereas the pilot cases are synthetic. The change of direction is partly a power effect and partly a distributional one, since real oncology correspondence leans more heavily on the MANAGE axis where the signal lives.
- **Geography drives it, not the name.** The ablation (Table 10) shows the geographic reference is 3 to 6 times stronger than the name as a perturbation channel. A patient's name on its own barely moves the metric, while their stated location does.
- **The signal is channel-specific.** On OncQA it sits on MANAGE and RESOURCE, not VISIT. The pattern fits a picture in which models are more willing to recommend home self-management, or to withhold a diagnostic referral, for a Global-South patient, while still correctly recommending an in-person visit when the case clearly warrants one.

RQ3 asks whether GDI yields a stable model ranking, and the evidence is mixed. Within the OncQA panel the ranking (Claude-Haiku-4.5 > GPT-4o-mini > Llama-3.3-70B) is consistent between the GDI point estimate and the Wilcoxon p-value, which is encouraging. But the ranking is *not* stable across panels and scales: GPT-4o-mini is the least-biased model in the pilot and the second-most-biased on OncQA. We therefore conclude that GDI is a usable pre-deployment comparator only when computed at adequate n on representative real data, and we do not claim a scale-invariant ranking.

7.3 Comparison to the Literature

Our OncQA per-model GDI (+0.009 to +0.045) and per-question RCER shifts (up to +10.8 pp on MANAGE) sit in roughly the same band reported for neighbouring identity axes. Gourabathina et al. [1] report tonal and gender RCER shifts of about 7 to 9 pp on GPT-4, and Omar et al. [16] report triage-decision shifts of 5 to 15 pp across race, housing, and LGBTQIA+ indicators. The geographic axis therefore produces disparities of similar size to the race and tone axes already documented in the literature. That is the central empirical claim of this project, and no prior clinical-LLM audit had measured it.

8 Baselines and Evaluation Methodology

A perturbation-based audit works differently from an accuracy benchmark, so its baselines are not the usual ones. This section states what they are and why they were chosen.

The primary baseline is within-subject and within-model. For each case c and model f_θ, the Global-North condition is the reference against which each Global-South condition is compared, and the analysis unit is the matched case-pair. This has three consequences. (i) Per-case variance in clinical complexity drops out as a confounder, because each case contributes a matched pair; a pooled analysis that ignored pairing would be noisier by roughly $\sqrt{2}$ on the same n. (ii) Per-model base-rate differences in recommendation style cancel, because the delta is computed within-model before any aggregation. (iii) The design measures *disparity*, not *accuracy*, so a model that is uniformly wrong under both conditions produces a near-zero GDI by construction.

What the baseline is not. A naive accuracy comparator that always recommends VISIT reaches 13/20 = 65% accuracy on the pilot's gold VISIT labels (the majority-class rates are 55% on MANAGE and 65% on RESOURCE). Under the matched-pair design this constant policy has GDI identically zero, because a constant policy cannot exhibit disparity, so it gives no discriminating information. Per-model accuracy against the gold labels is recorded for completeness but is not used as a decision rule anywhere in the pipeline.

External baselines. The TSR/RCR/RCER family was introduced by Gourabathina et al. [1] for tonal, gender, and syntactic perturbations. Omar et al. [16] and Pfohl et al. [9] supply per-axis effect-size bands for race, housing, LGBTQIA+, and demographic-parity comparisons. Our OncQA effects are set against these bands in §7.

Significance threshold. Two Bonferroni-corrected thresholds are used, for two purposes. The *descriptive* threshold $\alpha = 0.005$ ($= 0.05/9$ region $\times$ question conditions) is cited alongside Wilcoxon p-values and drives the power table. The *H1-vs-H2 discrimination* threshold corrects over the model $\times$ question tests; for the three-model OncQA panel this is $\alpha = 0.05/9 \approx 0.0056$. Per-model 95% BCa CIs are reported as descriptive, uncorrected effect-size summaries.

9 Threats to Validity and Limitations

We enumerate the principal threats to validity known to the authors, each with its mitigation or its status as an acknowledged limitation.

Gold-label provenance. Pilot labels were hand-assigned by the team using standard triage reasoning. A formal Emergency Severity Index v4 rubric [19] has been drafted (`code/docs/labeling_rubric.md`) and a Cohen's κ [18] computation script shipped (`code/scripts/compute_kappa.py`), but a board-certified-clinician double-labelling exercise was not completed within the project window and remains an acknowledged limitation. The OncQA cases are clinician-validated at source [20], which partly mitigates this for the headline experiment.

Annotator reliability. The Llama-3.1-8B annotator agreed with the authors on 38/40 (95%) spot-checked cases. A formal κ against independent human labellers was not completed and is a limitation; the heuristic-fallback rate in the final artefacts is 0.

Synthetic pilot dataset. The 20 pilot vignettes, while clinically realistic, do not capture the distributional quirks of real clinical text. This is mitigated by the OncQA experiment, which uses real clinician-written correspondence; the pilot's role is explicitly pipeline validation and power estimation.

Single seed. All runs use one seed. Multi-seed variance estimation was not performed and is the most important piece of future work for tightening the OncQA confidence intervals.

Panel migration. Provider rate limits forced the OncQA/ablation panel to differ from the pilot panel. This is why cross-table model comparison is explicitly disclaimed (§6.2); within-panel comparisons are unaffected.

Name-phonological confounds. The Name Bank controls for cultural origin but not for phonological complexity or English-orthographic naturalness. A confound-control regression is left to future work; the Name/Geo ablation partly addresses this by showing the name channel is the weaker one.

Geographic-reference confounds. Substituting "Boston, MA" with "Lahore, Pakistan" changes perceived origin and also real healthcare-system priors. The Geo-only ablation isolates the geographic channel but cannot separate "perceived identity" from "inferred health-system context"; disentangling these is future work.

Coverage. Two Name-Bank regions (Southeast Asia, MENA) were not exercised, and two planned real datasets (r/AskaDocs, USMLE+Derm) were not run within the project window. The reported findings are therefore scoped to OncQA oncology correspondence and three South regions.

Reproducibility. Every number traces to a timestamped run directory; the dataset and NameBank SHA-256 hashes are pinned in `manifest.json`; the cache keys every LLM call by SHA-256(model, prompt, seed, temperature), so any run reproduces byte-identically on the same config and seed.

10 Conclusion and Future Work

10.1 Conclusion

This project designed, built, and ran an end-to-end audit for geographic and cultural bias in clinical LLMs. The methodology is a matched-pair perturbation design with the TSR/RCR/RCER/GDI metric family, and it is the first to treat Global North versus Global South as a clinical perturbation axis. The implementation is a reproducible, near-zero-dependency, multi-provider pipeline, with every result pinned to a hashed run directory.

The empirical conclusion is that geographic-axis bias in clinical LLMs is genuine, but its size varies from model to model. A deliberately under-powered synthetic pilot returned a null. Scaling the same pipeline to 60 real OncQA oncology cases reversed that null: all three evaluated models showed a positive Geographic Disparity Index, and Claude-Haiku-4.5 crossed a Bonferroni-corrected significance threshold (GDI = +0.045, $p = 0.004$). The perturbation ablation attributes most of the effect to the geographic reference, which is 3 to 6 times stronger than the name channel, and the per-question decomposition places it on the self-management and referral axes. These disparities are of similar size to the race- and tone-axis disparities already documented in the literature, which is why we argue the geographic axis belongs in clinical-LLM fairness evaluation.

The research questions can now be answered directly. **RQ1:** on real data, Global-South identity perturbation does measurably reduce appropriate care. **RQ2:** the bias is systemic in direction, since all three models are positive on OncQA, but it varies in magnitude and significance from one model to the next. **RQ3:** GDI gives a usable within-panel ranking at an adequate n, but not a ranking that is stable across scales. **RQ4:** the disparity splits into a dominant geographic component and a weak name component, and the two do not simply add.

10.2 Future Work

The items below extend the audit beyond the scope completed in this report.

- **Additional real datasets.** Scale to r/AskaDocs verified- physician correspondence [21] and USMLE+Derm exam items [22, 10], whose gold distributions load less heavily on VISIT and would give an informative test on that axis.
- **Multi-seed variance estimation.** Re-run the variance-critical experiments under three or more seeds to tighten the BCa confidence intervals and confirm the OncQA significance.

- **Region coverage.** Exercise the Southeast Asia and MENA Name-Bank regions, already constructed, to test whether the disparity generalises beyond the three South regions reported here.
- **Intersectional analysis.** Cross the geographic axis with gender and disease severity to test for compounding effects.
- **Clinician inter-rater agreement.** Complete the board-certified-clinician double-labelling exercise and report Cohen's κ on the gold labels.
- **Mechanism, not just detection.** Run attention and activation analyses on the open-weights model (Llama-3.3-70B) to investigate *where* the geographic signal enters the computation.

11 Team Contributions

The project ran across four milestones: the proposal, the literature review, the intermediate report, and this final report. The table below records each member's primary ownership across that span. All members reviewed one another's code and report sections before each submission.

Member	Primary Contributions
Taimour Abdul Karim	Data and perturbation infrastructure: constructed and clinician-labelled the 20-vignette synthetic pilot dataset; led the Geographic Name-Bank v1.0 construction (150 entries $\times$ 6 regions, M/F balanced); built the dataset loader (`audit.data`) including OncQA vendoring and the gendered-cancer filter; co-designed and implemented the deterministic perturbation engine (`audit.perturb`); designed and calibrated the Llama-3.1-8B annotator (`audit.annotate`) with the three-layer fallback; authored the ESI v4 labelling rubric and the Cohen's κ script; wrote the Problem Formulation, Experimental Setup, Baselines, and Threats-to-Validity sections.
Syed Muhammad Mujtaba	Model inference, statistics, and reporting: built the unified multi-provider inference harness (`audit.models`) over OpenAI, Groq, and AWS Bedrock; implemented per-model token-bucket rate limiting and the Cloudflare User-Agent fix; led the Groq-to-Bedrock migration that unblocked the OncQA run; wrote the statistics module (`audit.metrics`) including the hand-rolled Wilcoxon, BCa bootstrap, and Cohen's h; wrote the power-analysis and figure-generation scripts and produced all five figures; executed the OncQA $n = 60$ run and the three ablation runs; integrated the LaTeX report and bibliography.
Usmar Haider	Literature grounding and manuscript quality: carried the Module 2 literature review forward, maintaining the four thematic clusters and the cross-milestone citation set; reviewed the Geographic Name-Bank for regional representativeness and M/F balance and verified entries against the manifest hash; edited the report prose end-to-end for clarity and consistency across the multi-author writing split; maintained the submission checklist against the LMS template; co-owns the Related Work section and the final presentation deck.

Member	Primary Contributions (continued)
Shawal Latif	Project framing and evaluation design: contributed to the Module 1 proposal (problem statement, expected outcomes) and the Module 2 literature review; helped design the GDI metric family and the matched-pair evaluation methodology; reviewed the statistical-analysis code and the significance-threshold (Bonferroni) treatment; contributed to the Analysis and Discussion section and proofread the final manuscript.
Abdul Moeed Irshad	Domain research and validation: contributed to the Module 1 proposal and the Module 2 literature review; researched the clinical triage rubric and gold-label conventions; assisted with case-vignette review and Name-Bank cultural-validity checks; contributed to the Conclusion and Future Work section and to cross-checking reported numbers against the run artefacts.

Table 11: Per-member primary contributions across the four project milestones.

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A Reproducibility and Run Inventory

Repository layout (the audit code ships with the Module 3 submission under `code/` and is unchanged for this report):

- `audit/data.py`: dataset loader and SHA-256 manifest hashing.
- `audit/perturb.py`: Name-Bank-driven deterministic perturbation engine.

- `audit/models.py`: unified `generate()` over OpenAI, Groq, and AWS Bedrock; per-model token-bucket rate limiter; idempotent disk cache.
- `audit/annotate.py`: Llama-3.1-8B annotator with three-layer JSON/regex/heuristic fallback.
- `audit/metrics.py`: TSR, RCR, RCER, GDI; paired Wilcoxon; BCa bootstrap; no SciPy dependency.
- `audit/run.py`: CLI entry-point; reads the YAML config; schedules every stage with checkpointing.
- `scripts/reannotate.py`, `scripts/power_analysis.py`, `scripts/ablation_compare.py`: re-annotation, power analysis, ablation aggregation.

Reproducing the runs. With `OPENAI_API_KEY` and AWS Bedrock credentials in `.env`:

```
set -a; source .env; set +a
cd Module_3_Intermediate_Report/code
# Synthetic pilot (OpenAI + Groq; requires GROQ_API_KEY)
python3 -m audit.run --config configs/pilot_oncqa.yaml --seed 42 --parallelism 8
# OncQA real-data scaling (OpenAI + AWS Bedrock)
python3 -m audit.run --config configs/oncqa_bedrock.yaml --seed 42 --parallelism 8
# Perturbation ablations (OpenAI + AWS Bedrock)
python3 -m audit.run --config configs/pilot_name_only_bedrock.yaml --seed 42 --parallelism 8
python3 -m audit.run --config configs/pilot_geo_only_bedrock.yaml --seed 42 --parallelism 8
```

Each run directory contains `manifest.json`, `perturbed.jsonl`, `completions.jsonl`, `annotated.jsonl`, `summaries.json`, and a SHA-256-keyed response cache. Re-running with the same config and seed produces byte-identical metrics. Every number in this report's tables traces to the `summaries.json` of the run directory cited in Table 4.

Run inventory. The reported results derive from five completed run directories: the synthetic pilot (`20260418T050306Z`), the OncQA $n = 60$ scaling run (`20260419T121941Z`), and three ablation runs (`20260419T123259Z` Name-only, `20260419T123612Z` Geo-only, `20260419T123954Z` Combined). Aggregate ablation figures are in `runs/ablation_summary.json`.